MIL-C-48476B(AR)
30 June 1981
SUPERSEDING
MIL-C-48476A(AR)
20 July 1976

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11732372 WIND ELECTRONICS

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11732372 Wind Electronics which is the A9 component of the M21 Ballistic Computer Unit, 11732400.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Maintainance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11732372	Circuit Card Assembly: 11732372 Wind Electronics

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750426	Adapter, Computer Board - A9
11821457	Fixture Shock, Vertical and Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
11821500	Vibration/Shock, Test Fixture - Computer Unit
11821520	Automatic Test Assembly Central Processor Unit
11821854	Test Fixture Auto Card, Test Set, Computer, FAT

Packaging Data Sheet:

P11732372	Packaging of Circuit Card Assembly: 11732372

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The wind electronics circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11732372 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1).

3.1.1 Signals. The assembly shall provide the following signals:

Signal Designation

a. TRIM 2
b. V_w
c. WIND SENSOR FAIL
d. POWER OK
e. BRIDGE 1
f. WG (WIND GOOD)

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{o} \pm 18^{o}F$
b.	Relative humidity	50 percent + 30 percent
c.	Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 TRIM 2 and V_w outputs. With the loads, interconnections, and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 2, 4.1, 4.3, 4.5, 4.6 and 4.7 applied and with the input signals (table II) applied, the outputs at PI-17B (TRIM 2) and PI-17A (V_w) shall be as specified in table II. (see 4.6.2.1).

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3.3.2 V_W settling time. With the conditions of 3.3.1, table II, item 1 having been stabilized at least 30 seconds, and then the input at PI-22A having been switched to the condition of item 6, the voltage at PI-17A shall settle to 1.50 $\pm$ 0.50 VDC within 15 $\pm$ 4 seconds after the input was switched. (see 4.6.2.2)

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TABLE 1. Loads, power forms, signals and interconnections.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected between pin P1-A11 and 24 VDC (see item 2.2)
1.1	Resistor	510 Ohm ± 5%, 2W	
1.2	Network	Figure 1	Connected as shown on figure 1
1.3	Network	Figure 2	Connected as shown on figure 2
1.4	Resistor	3.67k ± 0.5%	Connected between pin P1-B24 and P1-B17
1.5	Resistor	3.35k ± 1%	Connected between pin P1-A24 and P1-A18
1.6	Network	Figure 3	Connected as shown in figure 3
1.7	Resistor	1 MEG ± 5%	Connected between pin P1-A17 and P1-A18
2.0	Power Sources		
2.1	30 VDC	32.5 ± 2.5 VDC Ripple 0.5 V p-p max	P1-1A, return to P1-3A
2.2	24 VDC	Tolerance: ± 2 VDC Ripple 1.0 V p-p max	Resistor (item 1.1), return to P1-3A
2.3	15 VDC	Tolerance: ± 0.030 VDC Ripple 0.020 V p-p max	P1-6A, return to P1-18A
2.4	10 VDC	Tolerance: ± 0.030 VDC Ripple 0.010 V p-p max	P1-19A, return to P1-18A
2.5	-15 VDC	Tolerance: ± 0.030 VDC Ripple 0.020 V p-p max	P1-21A, return to P1-18A

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TABLE I. Loads, power forms, signals and interconnections. - Continued

Item	Conditions	Characteristics	Connections
3.0	Signal source		
3.1	Selectable: -10 to 10 VDC	Tolerance as specified Ripple 0.010 V p-p max	As required, return to P1-18A
3.2	Selectable: -10 to 10 VDC	Tolerance as specified Ripple 0.010 V p-p max	As required, return to P1-18A
4.0	Interconnections		
4.1	Ground	Jumper wire	P1-A3, P1-A13, and P1-A18
4.2	BIT	+15.0 ± 2.0 VDC 1/ -15.0 ± 2.0 VDC 2/	applied to P1-A16 applied to P1-A15
4.3	$\overline{\text{BIT}}$	-15.0 ± 2.0 VDC 2/ +15.0 ± 2.0 VDC 1/	applied to P1-A16 applied to P1-A15
4.4	MANUAL WIND	+15.0 ± 2.0 VDC 1/ -15.0 ± 2.0 VDC 2/	applied to P1-B15 applied to P1-B14
4.5	$\overline{\text{MANUAL WIND}}$	-15.0 ± 2.0 VDC 2/ +15.0 ± 2.0 VDC 1/	applied to P1-B15 applied to P1-B14
4.6	BRIDGE 2	+3.30 ± 0.01 VDC	applied to P1-A2
4.7	PWR OK	+10.00 ± 0.50 VDC	applied to P1-A7
4.8	$\overline{\text{PWR OK}}$	0.00 ± 0.50 VDC	applied to P1-A7

1/ Source impedance shall be 510 ohms ± 5%, 1/2 w.
2/ Source impedance shall be 10K to 20K ohms.

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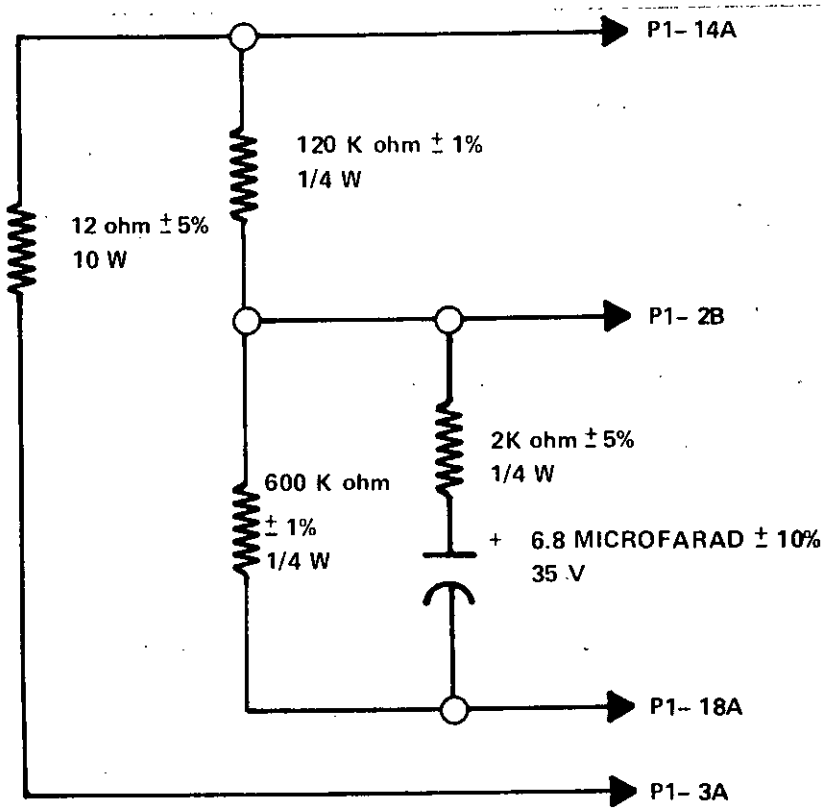


FIGURE 1. Functioning probe load circuit.

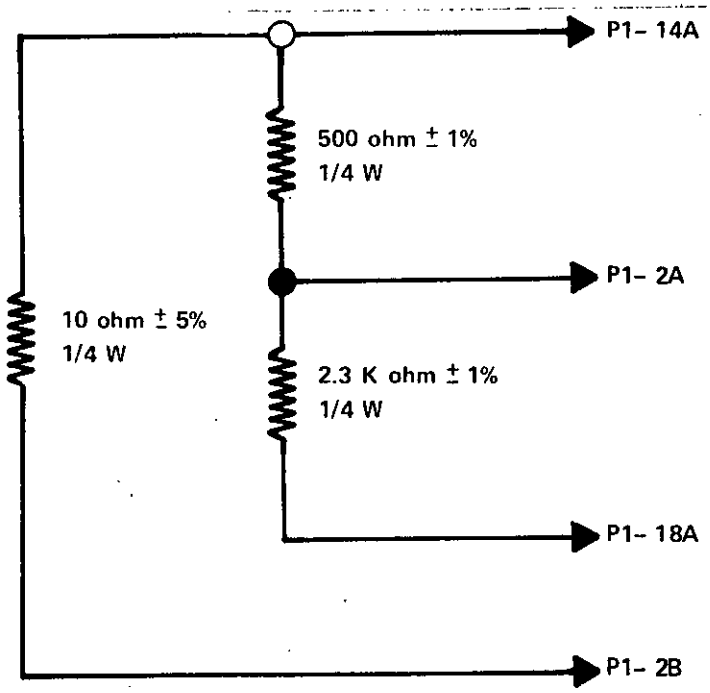


FIGURE 2. Broken probe load circuit.

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TABLE II. TRIM 2 and V_W characteristics.

Item	P1-22A 1/	P1-25B 2/	P1-25A 3/	Outputs (vdc)(3)	
				P1-17B	P1-17A 4/
1	-9.140	0	+15	$(-9.140 - V_{22A}) * 2.220 +7.9954 \pm 0.0775$	$(9.140-V_{22A}) * 2.220 -8.0444 \pm 0.3265$
2	-8.230	0	+15	$(-8.230 - V_{22A}) * 2.220 +6.0065 \pm 0.1171$	Not applicable
3	-7.295	0	+15	$(-7.295 - V_{22A}) * 2.220 +3.9964 \pm 0.1533$	Not applicable
4	-5.790	0	+15	$(-5.790 - V_{22A}) * 0.836 +1.9925 \pm 0.1294$	Not applicable
5	-4.250	0	+15	$(4.250 - V_{22A}) * 0.400 +0.9898 \pm 0.0551$	Not applicable
6	0.000	0	+15	$(0.000 - V_{22A}) * 0.100 +0.0000 \pm 0.0150$	Not applicable
7	0.000	+15	0	$-(0.000 - V_{22A}) * 0.100 -0.0006 \pm 0.0150$	$-(0.000 -V_{22A}) * 0.100 +0.0003 \pm 0.0508$
8	-4.250	+15	0	$(4.250 - V_{22A}) * 0.400 -0.9892 \pm 0.0760$	Not applicable
9	-5.790	+15	0	$(5.790 - V_{22A}) * 0.836 -1.9919 \pm 0.1707$	Not applicable
10	-7.295	+15	0	$(7.295 - V_{22A}) * 2.220 -3.9958 \pm 0.2360$	Not applicable
11	-8.230	+15	0	$(8.230 - V_{22A}) * 2.220 -6.0059 \pm 0.2413$	Not applicable
12	-9.140	+15	0	$(9.140 - V_{22A}) * 2.220 -7.9949 \pm 0.2428$	$(9.140 -V_{22A}) * 2.220 +8.0471 \pm 0.4948$

1/ Tolerance = 0.010 vdc (item 3.1, table I)
2/ +15 = item 2.3, table I; 0 = return of item 2.3, table I
3/ V_{22A} is actual value of input at P1-22A.
4/ Allow at least 45 seconds after application of inputs for voltage to settle.

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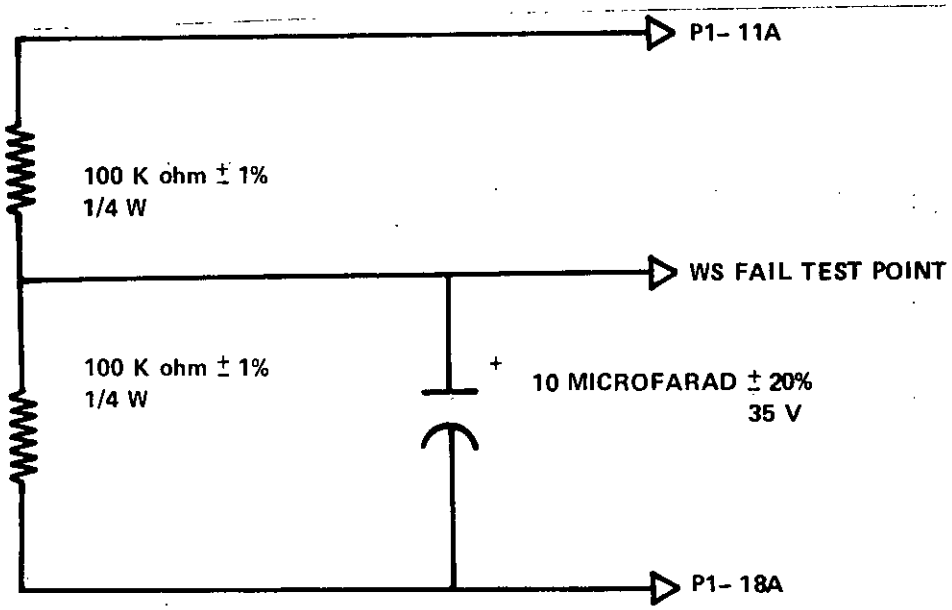


FIGURE 3. Load circuit (WS FAIL TEST POINT).

3.3.3 V_W (BIT) output. With the loads, interconnections, and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 1.7, 2, 4.1, 4.2, 4.6 and 4.7) applied and with 0.000 ± 0.010 VDC (item 3.1) applied at P1-22A, 0.000 ± 0.010 VDC applied at P1-16B, 0 VDC applied at P1-25A and +15 VDC applied at P1-25B, the output at P1-17A (V_W) shall be 4.113 ± 0.060 VDC under each of the following:

a. With interconnection (table I, item 4.4) applied.
b. With interconnection (table I, item 4.5) applied and allowing 15 seconds for the output to settle. (See 4.6.2.3)

3.3.4 V_W (MANUAL WIND) output. With the loads, interconnections and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 2, 4.1, 4.3, 4.4, 4.6 and 4.7) applied and with the 8.0 ± 0.5 VDC (item 3.1) applied at P1-16B; the output at P1-17A shall be 8.00 ± 0.50 VDC. With the table I, item 4.4 signal source removed and the table I, item 4.5 signal source applied and with ground applied at P1-22A the output at P1-17A shall be 0.00 ± 0.10 VDC. (See 4.6.2.4)

3.3.5 WIND SENSOR FAIL output. With the loads, interconnections, and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 2. 4.1, 4.3, 4.5, 4.6, and 4.7) applied, the voltage at WS FAIL TEST POINT, figure 3, shall be 12.0 ± 1.0 VDC. The voltage at WS FAIL TEST POINT shall be as specified with the following conditions. Allow at least 10 seconds for voltage to settle.

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a. WS FAIL TEST POINT voltage shall be less than 2.0 VDC when the 30 VDC power (table I, item 2.1) is removed.

b. WS FAIL TEST POINT voltage shall be less than 7.5 VDC when the load (table I, item 1.2) is removed.

c. WS FAIL TEST POINT voltage shall be less than 2.0 VDC when the functioning probe load (table I, item 1.2) is removed and the broken probe load (table I, item 1.3) is applied. (See 4.6.2.5).

3.3.6 POWER OK output. With the loads, interconnections, and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 2, 4.1, 4.3, 4.5, 4.6, and 4.7) applied, the voltage at WS FAIL TEST POINT, figure 3, shall be 12.0 ± 1.0 VDC. With the table I, item 4.7 signal source removed and the table I, item 4.8 signal source applied, the voltage at WS FAIL TEST POINT shall be less than 2.0 VDC. Allow not less than 10 seconds for voltage at WS FAIL TEST POINT to settle. (See 4.6.2.6)

3.3.7 BRIDGE 1 output. With the loads, interconnections and power (table 1, items 1.1, 1.2, 1.4, 1.5, 1.6, 2, 4.1, 4.3, 4.5, 4.6, and 4.7 applied, and -9.140 ± 0.010 VDC (table I, item 3.1) applied to P1-22A, the voltage difference between P1-2A and P1-2B shall be 0.000 ± 0.003 VDC and the voltage at P1-14A shall be 4.00 ± 0.15 VDC with a ripple content of less than 0.150 Vrms. Allow not greater than 12 seconds after application of inputs for output voltages to settle. (See 4.6.2.7)

3.3.8 Direction comparison signal. With the loads, interconnections, and power (table I, items 1.1, 1.2, 1.4, 1.5, 1.6, 2, 4.1, 4.3, 4.5, 4.6 and 4.7 applied, and with the inputs of table III applied, the outputs at P1-17B and TP10 shall be as specified in table III. (See 4.6.2.8)

3.3.9 WG output. With the loads, interconnections and power (table I, items 1.1, 1.4, 1..5, 1.6, 1.7, 2, 4.1, 4.3, 4.6 and 4.7) applied and with -9.140 ± 0.010 VDC (item 3.1) applied at P1-22A the output at P1-17A (V_W) shall be as specified, allowing not less than 45 seconds after application of inputs for the voltage to settle, with the following conditions:

a. With load and interconnection (table I, items 1.2 and 4.4) applied and with +15 VDC applied at P1-25A and 0 VDC applied at P1-25B the voltage at P1-17A (V_W) shall be 0.00 ± 0.05 VDC (Rtn at P1-18A).

b. With load and interconnection (table I, items 1.3 and 4.5) applied and with 0 VDC applied at P1-25A and +15 VDC applied at P1-25B the voltage at P1-17A (V_W) shall switch from 0.0 + 0.1 VDC to 8.0 ± 1.0 VDC not less than two times a second. (See 4.6.2.9)

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TABLE III. Direction comparison signal.

Item	Inputs (VDC) 1/		Outputs	
	P1-25A	P1-25B	P1-17B	TP10 (VDC)
1	3.000 ± 0.003	1.600 ± 0.002	-8.00 ± 0.50 VDC	-1.000 ± 0.220
2	3.000 ± 0.003	1.500 ± 0.002	Not applicable	0.000 ± 0.220
3	3.00 ± 0.003	1.400 ± 0.002	+8.00 ± 0.50 VDC	1.000 ± 0.220
4	0.00 ± 0.002	0.000 ± 0.002	Square wave 2/	0.000 ± 0.045

1/ Items 3.1 and 3.2, Table I.
2/ Period equal 88.4 ± 18.6 microseconds, no rise time requirement.

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 4. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4, 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4, 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

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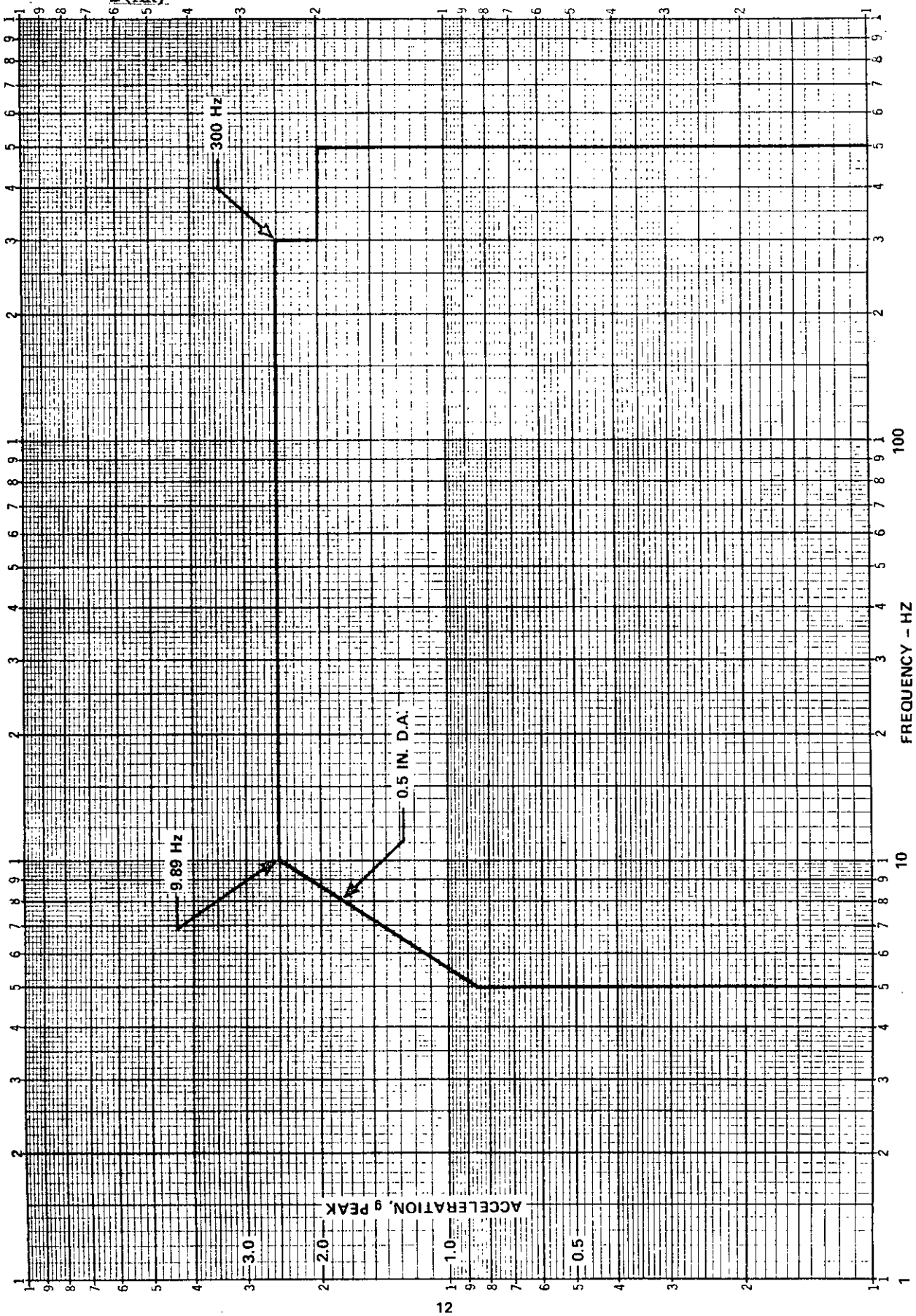


FIGURE 4. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classificaaion of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables IV, V and VI.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE IV. Performance tests.

Item	Characteristic	Requirement	Test Procedures
101	TRIM 2 and V_w outputs	3.3.1	4.6.2.1
102	V_w settling time	3.3.2	4.6.2.2
103	V_w (BIT) output	3.3.3	4.6.2.3
104	V_w (MANUAL WIND) output	3.3.4	4.6.2.4
105	WIND SENSOR FAIL output	3.3.5	4.6.2.5
106	POWER OK output	3.3.6	4.6.2.6
107	BRIDGE 1 output	3.3.7	4.6.2.7
108	Direction comparison signal	3.3.8	4.6.2.8
109	WG output	3.3.9	4.6.2.9

TABLE V. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE VI. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

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4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected aa random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table V.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VI shall be applied separately to each assembly in the sample.

4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table V and only the test of item 307 in table VI shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table IV and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

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4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732372.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table VII shall be used where specified to perform required tests.

TABLE VII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750426	Adapter, Computer Board - A9
3	11821457	Fixture Shock, Vertical and Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821500	Vibration/Shock Test Fixture - Computer Unit
6	11821520	Automatic Test Assembly Central Processor Unit
7	11821854	Test Fixture Auto Card, Test Set, Computer, FAT

4.6.2 Performance tests.

4.6.2.1 TRIM 2 and V_w outputs. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.1.

4.6.2.2 V_w settling time. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.2.

4.6.2.3 V_w (BIT) output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.3.

4.6.2.4 V_w (MANUAL WIND) output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.4.

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4.6.2.5 WIND SENSOR FAIL output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.5.

4.6.2.6 POWER OK output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.6.

4.6.2.7 BRIDGE 1 output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.7.

4.6.2.8 Direction comparison signal. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.8.

4.6.2.9 WG output. Use items 1, 2 and 6 of table VII or suitable equipment to verify the requirements of 3.3.9.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 5 of table VII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 + g with a time duration of 18 + 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.1.2 High intensity. Use items 3, 4 and 5 of table VII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half since wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 + 10.0 g with a time duration of 1.5 + 0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IV.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 4 and 5 of table VII or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration specified curve in figure 4 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table IV.

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4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table VII or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table IV shall be applied.

4.6.3.4 Low temperature. Use item 7 of table VII or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table IV shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 5. At the conclusion of this test, subject the assembly to the performance tests of table IV.

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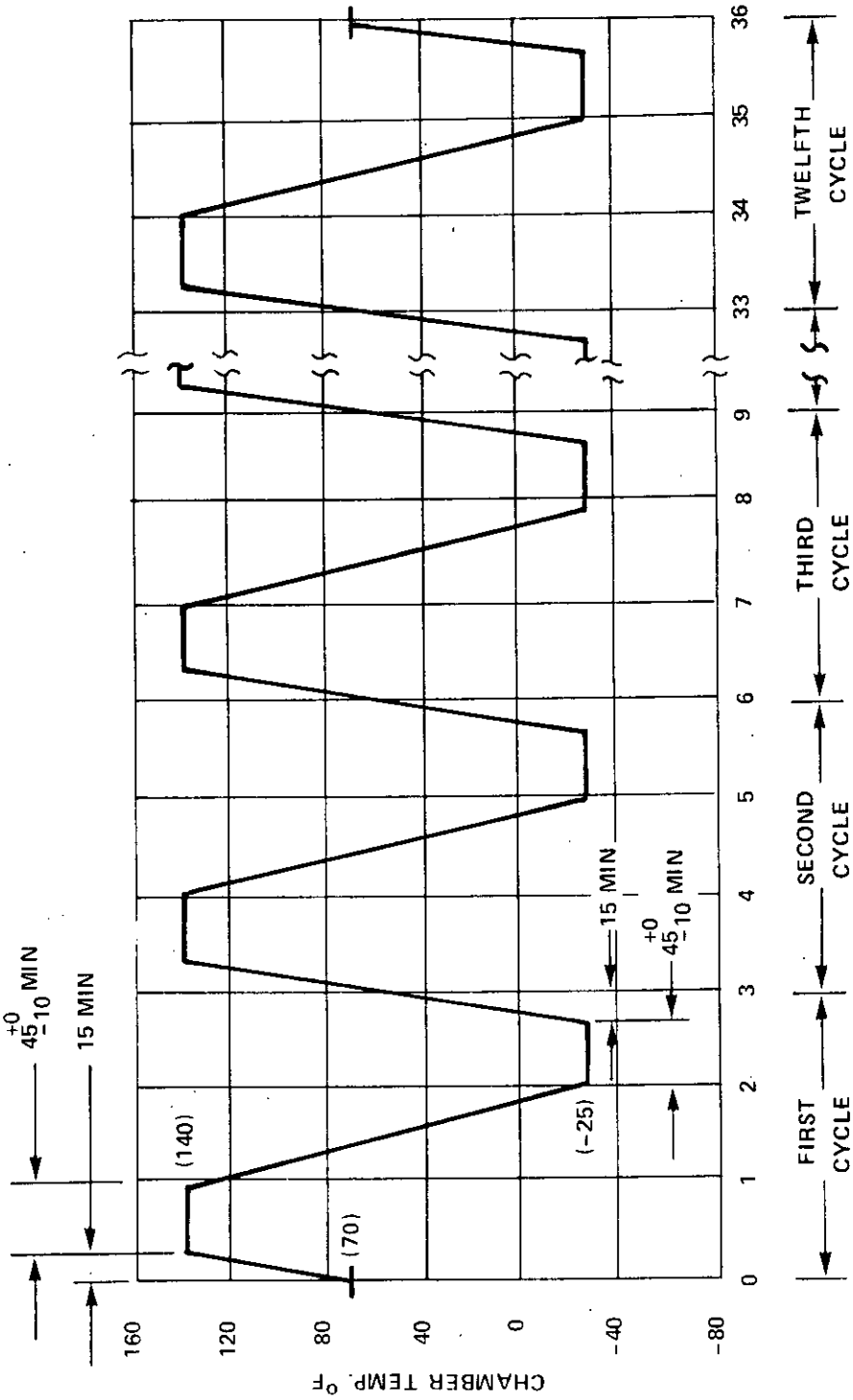


FIGURE 5. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732372. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Wind Electronics circuit card assembly is used in the M21 Ballistic Computer Unit.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required, and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging, and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48476B was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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MIL-C-48476B CIRCUIT CARD ASSEMBLY; 11732372 WIND ELECTRONICS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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